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Analysis of electro-thermo-mechanical behavior of thin film Ni_{50}-Ti_{50} and Ni_{40}-Ti_{50}-Cu_{10} shape memory alloys for application in thermal actuators

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ABSTRACT

The article demonstrates the electrothermomechanical behavior of thin-film for two different shape memory alloys by the Finite Element analysis method. The effect of varying input currents and different thin film thicknesses on the stress, temperature, and deformation is analyzed. Comparing the results obtained for the two alloys concludes that Ni_{40}-Ti_{50}-Cu_{10} thin film is a better choice for actuator applications. The results presented can assist in preparing a predictive model, which further helps forecast the corresponding values of electric current and film thickness for the desired temperature and displacement. Controlling the desired temperature based on the inputs provided can be a better method of programming and actuating the SMA thin film using Joule's heating mechanism.
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1. Introduction

During recent years, MEMS (also known as microsystems) has thrived to enter various applications, including automobile, aerospace, consumer, and biomedical equipments (see Fig. 1). MEMS devices are the amalgamation of sensors, electrical connections, mechanical components, and actuators within a standard device. This process employs microfabrication and micromachining techniques for manufacturing [1]. Advantages of using microsystems are the elimination of consumables (very few additives in lab on a chip device), quicker reaction time (automobile airbag sensors), improved compactness, more excellent resolution (print head of the inkjet printers), etc. [2]. Various industries for better and efficient microsystem devices need more significant effort from material science technology to develop new alloys and compounds.

Ni-Ti alloy's accidental discovery in the year 1959 paved the way for using shape memory alloys (SMA) for different applications. Various SMAs have been intensively analyzed to increase the industry relevance of shape memory technologies, such as magnetic SMAs, high-temperature SMAs, etc. The physical basis of the shape memory effect (SME) in SMA materials is because of a diffusion-free martensitic shift between the higher temperature phase (austenite) and the low-temperature phase (martensite) [3]. A plateau and hysteresis are typical for the transition when the load is removed. The magnitude of the pseudoelastic/ superelastic strain for single crystals can be as large as 8% or more. Latest advances in self-actuated SMA components with a feature scale of few nanometers reveal that the SME still prevails down to nanometer dimensions as long as the creation of twin boundary is energetically favored. Besides, the down-scaling of the structure is very useful in terms of the intricacies of heat transfer. Another intriguing characteristic of SMA material is its multifunctionality, which allows for structural stability, electrical and thermal conductivity, actuation, and self-sensing at about the same time [4]. In this sense, the concept of "the material is the machine" opens up new routes for developing small-scale smart architectures that can be fabricated, which were not feasible before.

With the advancement of technology and new fabrication processes, these SMA materials have been used in MEMS in the last decade. Smart materials like Piezoelectric materials, shape memory alloys, and ionic polymer composites have gathered attention to microactuators' expansion. The most crucial advantage of SMA over other smart materials is maximum energy density, which

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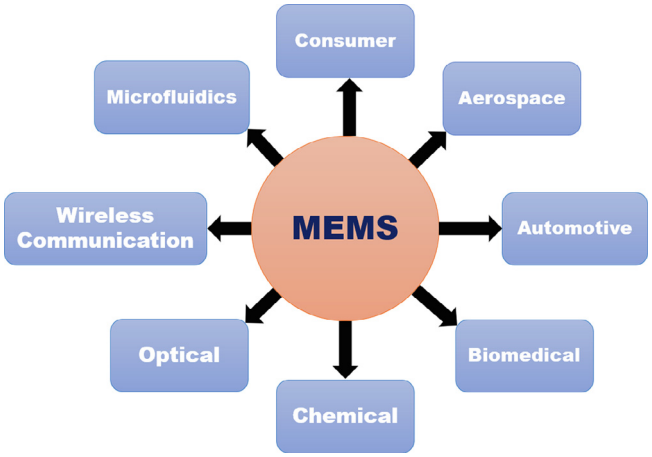


Fig. 1. A broad spectrum of application of MEMS (Microsystems).

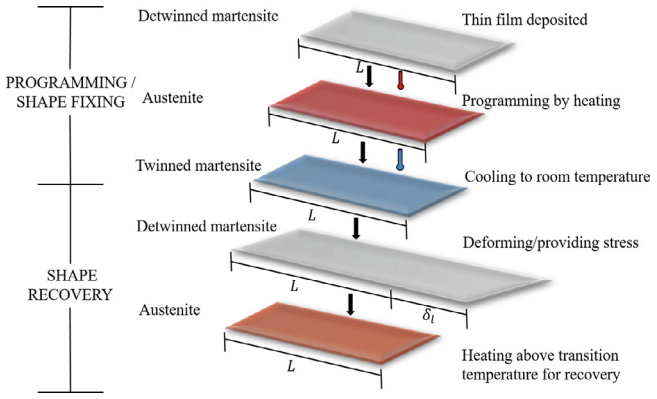


Fig. 2. Utilizing Joule's heating for programming and recovery in shape memory alloys.

can be very useful for high-performance microactuators. SMA's unique ability to recover enormous strain is of utmost importance in advancing micro-scale devices, considering micropumps, microgripper, microcantilever, micro thermostat arrays, and micromirrors. The ability to fabricate SMA in thin films has been a critical enabler for its uses like sensors, actuators, and manipulators in the medical application, microsystem, defense sector, and aerospace [5]. These can operate in very harsh external environmental conditions like highly radioactive areas, corrosive conditions, and space. Some examples of the SMA-MEMS devices reported by various researchers are micro manipulating devices [6], microvalves [7,8], micro pump [9–12], Thermo actuated microendoscopic devices [13], micro grippers [14–17], etc.

Researchers have utilized finite element simulations to analyze unimorph/ bimorph microactuator's deformation behaviors to reduce the experimental iterations, fabrication cost, time, and material wastage. The studies are carried out in several situations by adjusting the SMA thickness, stress layers, and deposition temperature. The same is also validated experimentally. The outcomes revealed that when the SMA film thickness was reduced, the displacement of the SMA microactuator increased [18]. Researchers have used commercial softwares, which includes a definable shape memory effect to show the Ni-Ti recrystallization, followed by thermal cycling. The experimental results were compared with the simulated results, and a fair consensus is observed in terms of temperature versus tip deflection [19]. Similarly, others have also utilized finite element analysis for different actuator systems involving SMAs [20–22]. Although some quality research in this field has been done earlier, electro-thermo-mechanical analysis using the FEA method has not been reported hitherto for Ni_{50}-Ti_{50} and Ni_{40}-Ti_{50}-Cu_{10} thin films. There is no predictive model that can be used as a guide map for programming and recovery of shape memory alloy thin films using joule's heating to the best of the authors' knowledge. In this work, an attempt has been made to analyze the actuation characteristics of the specific SMA thin films using FEA. A comparative study is presented between two different Ni-Ti-based SMA thin film compositions as per the FE analysis results. The effect of varying input current and thin film thickness influencing the displacement, stress, and temperature is also analyzed.

2. Methodology

The enablers of the shape memory effect are the programming temperature and recovery temperature as depicted in Fig. 2. There are different processes through which the programming and actuation heat can be provided. The heat ought to be above the transformation temperature of the SMA. One of the famous processes is by Joule's heating. Joule heating (also called ohms or resistive heating) defines the mechanism of converting the electrical energy into heat as it passes across a resistance. Electric power is transformed into heat through resistant material losses as the electric current passes through a solid or liquid with minimal conductivity. As the electrons transfer energy into atoms by means of collisions, the heat is produced in microscale. The actuation principle is due to the thermal expansion in the SMA material from the heat-induced in the conductor's current.

3. Finite element analysis based on joule's heating for two different SMAs

3.1. Model definition

The thin film, along with the air envelop in the form of a cuboid around it, is modeled using the geometry section of the model definition window provided in the FE simulation software. For the initial analysis, the length and breadth of the thin film are kept constant at 45 μm and 105 μm, respectively. The thickness of the film is varied from 0.5 μm to 2 μm. The dimension of the air envelope is 100 μm (length), 150 μm (breadth), and 50 μm (height), as shown in Fig. 3.

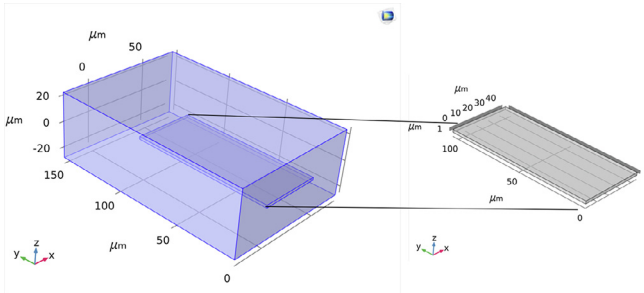


Fig. 3. 3D model of the thin film and the air envelop.

3.2. Material selection

There are two materials chosen for the present study. The first one is from the material library Ni-Ti (Nitinol), and other material Ni-Ti-Cu is added as a custom material. All the properties of the materials are taken from various literature which is provided in Table 1.

Table 1
Properties of SMAs considered from various literature [23–26].

Properties	Ni_{50}-Ti_{50}	Ni_{40}-Ti_{50}-Cu_{10}
Young's modulus (GPa)	55	35
Poisson's ratio	0.3	0.33
Density (kg/m^3)	6500	6500
Thermal Conductivity (W/(m.K))	18	18
Heat Capacity at constant pressure (J/kg.K)	400	460
Electrical Conductivity (S/m)	1219512	1666666.6
Relative Permittivity	1	1
Coefficient of thermal expansion (1/K)	10x10^{-6}	15.6x10^{-6}

3.3. Boundary conditions

One end of the thin film is fixed, and the remaining part is chosen to be free. The conductive heat transfer is considered in solids, and the initial temperature is assumed to be 293.15 K. The initial impedance (Z_{ref}) chosen is 50 Ω, and the electric potential is considered as 0 V. The fixed end is considered grounded, and the other open end is the terminal provided with the variable current starting from 2 mA to 18 mA. There is convective heat transfer chosen amongst the thin film of SMA and the nearby moist air. The boundary conditions considered are represnted in the Fig. 4. The properties of the air is directly taken from the materials library of the software package.

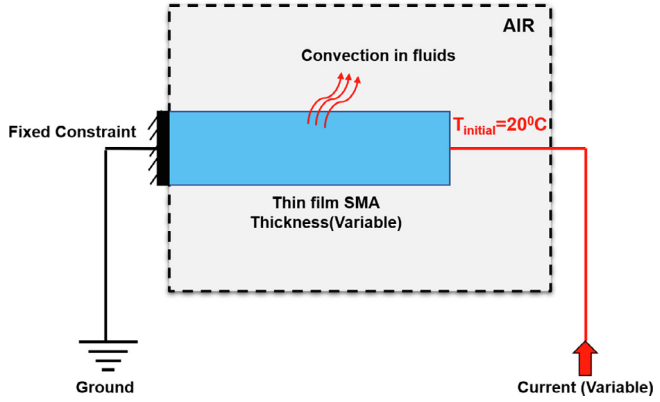


Fig. 4. Model for the FEA analysis with boundary conditions.

3.4. Governing equations and multiphysics considered

3.4.1. Electric current

The electric current module is used to compute electric field, current, and potential distributions in conducting media. It solves a current conservation equation based on Ohm's law using the scalar electric potential as the dependent variable. The differential form of Joule's heating is given by equation (1). Electric current is coupled with heat transfer in solids.

$$Q = J.E \tag{1}$$

where J is current density & E is the electric field. The values of J and E is calculated from equations (2) and (3), respectively.

$$J = \sigma E \tag{2}$$

$$E = -\nabla V \tag{3}$$

where σ is the electrical conductivity of the material and V is the electric potential.

3.4.2. Heat transfer

The temperature equation defined in solid domains corresponds to the Fourier's law of heat conduction (Equation (4)). Newton's law of cooling governs the convective heat transfer in between the film and the moist air (Equation (6)). Heat transfer in solids is coupled with solid mechanics.

$$\varepsilon T = \alpha(\Delta T) \tag{4}$$

where εT is thermal strain, α is coefficient of thermal expansion, and ΔT denotes the temperature change. The value of T is computed from equation (5).

$$q = -K\nabla T \tag{5}$$

where q is the local heat flux density, K is the thermal conductivity of the material, and ∇T is the temperature gradient.

$$\frac{dT}{dt} = k(M - T) \tag{6}$$

where T is the temperature of an object, M is the temperature of its surroundings, k is the heat transfer coefficient, and t denotes time.

3.4.3. Structural mechanics

The solid mechanics module is intended for general structural analysis of 3D, 2D, or axisymmetric bodies. It is based on solving the equations of motion together with a constitutive model for solid material (Equation (7)). Results such as displacements, stresses, and strains are computed.

$$0 = \nabla.S + F_V \tag{7}$$

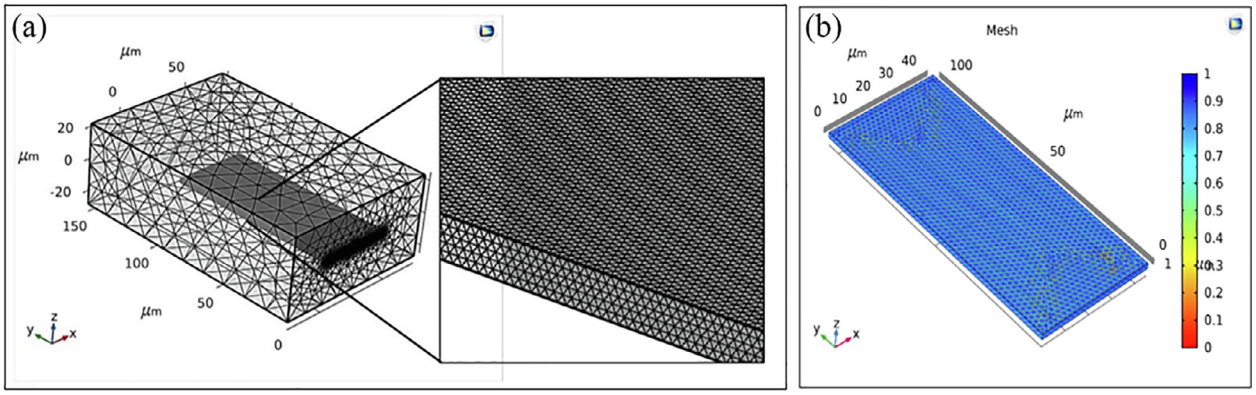


Fig. 5. (a) Meshing of the geometry; (b) Mesh quality analysis.

where S is stress tensor and F_V is the volume force vector. The value of S is computed by using equations (8) and (9).

$$S = \mathbb{C} : \epsilon_{el}, \epsilon_{el} = \epsilon - \epsilon_{inel} \tag{8}$$

$$\epsilon = \frac{1}{2}\left[(\nabla u)^T + \nabla u\right] \tag{9}$$

where $\mathbb{C}$ is the fourth-order stiffness tensor, u is displacement vector, $\epsilon_{el}$is elastic strain, ϵ_{inel} is inelastic strain and ϵ is the total strain.

3.5. Meshing

A fine tetrahedral mesh is generated. The full mesh consists of 54,289 domain elements, 6808 boundary elements, and 359 edge elements (see Fig. 5 (a)).

3.5.1. Mesh quality analysis

It is necessary to have a mesh quality analysis to get a mesh that will fetch accurate results and incur a low computational cost. The mesh quality analysis done for the present study is based on skewness as per equation (10).

$$Skewness = \frac{Optimal\ cell\ size - Cell\ size}{Optimal\ cell\ size} \tag{10}$$

Quality is defined as the inverse of the skewness. The average mesh quality is found to be more than 80%, which is acceptable (see Fig. 5 (b)).

4. Results and discussion

Using the MEMS Joule's heating module, the electro-thermo-mechanical behavior of the two different compositions of SMA is analyzed. The current and the thickness are varied, and its effect on the temperature generated, displacement, and thermal stresses are computed. Fig. 6 shows the FE analysis result for 0.5 μm thick Ni-Ti sheet for different electric current. Fig. 7 show the FE analysis result for 0.5 μm thick Ni-Ti-Cu sheet for different currents. Similarly, for the remaining three thicknesses, the results are computed and are presented in Table 2. The details of the result considering each of the thicknesses and various currents for both the SMAs (Ni_{50}-Ti_{50} and Ni_{40}-Ti_{50}-Cu_{10}) are presented in the comparative analysis section.

4.1. Results for Ni_{50}-Ti_{50}

After providing model definition, input material properties, and boundary conditions, the governing equations for the Multiphysics problem are used to compute surface von Mises stress, the temperature generated, and the total displacement for further analysis. In Fig. 6 (a), there is an increase in thermal stresses in the Ni-Ti thin film with an increase in current. The predicted stress value ranges from a minimum of $5.7x10^5$ N/m^2 to $4.7x10^7$ N/m^2. The cause of the increase in thermal stresses is the direct proportionality of input current to the heat generated in the thin film. During the convective heat transfer, the thermal stresses caused are more. In Fig. 6 (b), As current is proportional to the heat generated in the thin film, there is an increase in temperature with an increase in input current. The range of current is from 21.7 °C to 151 °C. Different values of temperature corresponding to different values of current can be helpful in decision making for choosing the exact value of current as per the required temperature for the end-user. In Fig. 6 (c), with an increase in current, there is an increase in the total displacement of the thin film. The range of displacement generated in the thin film ranges from a few nanometer to 100 nm. The actuation can be controlled by changing the current using Joule's heating. This nano/micro range actuation maintained precisely with changing input current can be advantageous for thermal actuator applications.

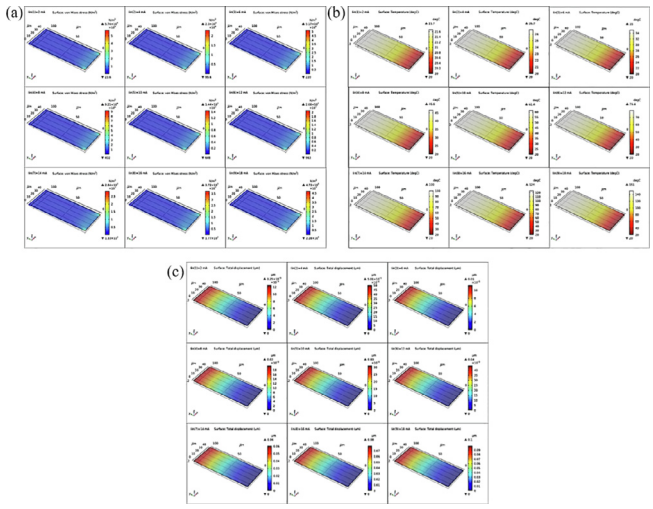


Fig. 6. Results for 0.5 μm thin-film Ni_{50}-Ti_{50} (a) von Mises stress; (b) Temperature generated; (c) Total displacement.

4.2. Results for Ni_{40}-Ti_{50}-Cu_{10}

Everything remains the same; as per the change in material properties for Ni-Ti-Cu, the results are computed and further analyzed. In Fig. 7 (a), there is an increase in thermal stress with an increase in electric current. The value of stress is ranging from $6.5x10^5$ N/m^2 to $5.3x10^7$ N/m^2. In Fig. 7 (b), there is an increase in temperature generated with an increase in current, and the range of current obtained is from 21.2 °C to 117 °C. In Fig. 7 (c), with an increase in the input electric current, there is an increase in the total displacement.

For both Ni-Ti & Ni-T-Cu, it has been found that there is an increase in stress with an increase in input current. The temperature increased as the input current increased and decreased as the film thickness increased. The displacement increased with an increase in current and decreased with an increase in the film thickness.

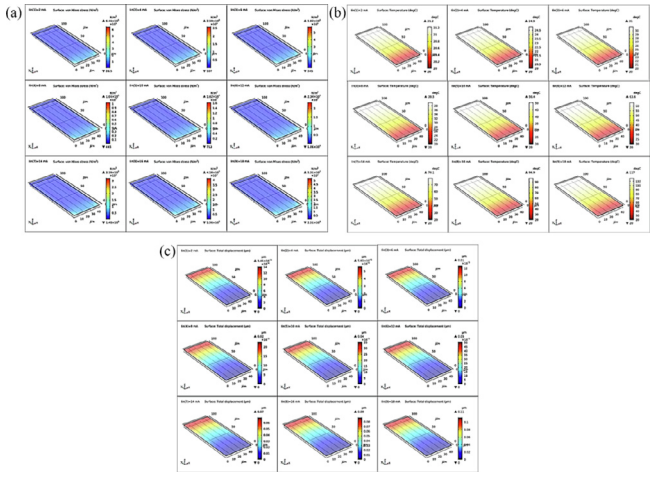


Fig. 7. Results for 0.5 μm thin-film Ni_{40}-Ti_{50}-Cu_{10} (a) von Mises stress; (b) Temperature generated; (c) Total displacement.

Table 2
Computed results of stress, temperature, and displacement in two selected SMAs for different values of current and film thickness.

0.5 μm thickness							1 μm thickness					
Ni_{50}-Ti_{50}				Ni_{40}-Ti_{50}-Cu_{10}			Ni_{50}-Ti_{50}			Ni_{40}-Ti_{50}-Cu_{10}		
I (mA)	Stress (N/m^2)	T (°C)	T.D (μm)	Stress (N/m^2)	T (°C)	T.D (μm)	Stress (N/m^2)	T (°C)	T.D (μm)	Stress (N/m^2)	T (°C)	T.D (μm)
2	5.7E + 5	21.7	1.3E-3	6.5E + 5	21.2	1.4E-3	1.3E + 5	20.5	3.4E-4	1.5E + 5	20.3	3.8E-4
4	2.3E + 6	26.7	5E-3	2.6E + 6	24.9	5.7E-3	5.3E + 5	21.8	1.4E-3	6E + 5	21.3	1.5E-3
6	5.2E + 6	35	0.01	5.8E + 6	31	0.01	1.2E + 6	24.1	3E-3	1.4E + 6	23	3.4E-3
8	9.2E + 6	46.6	0.02	1E + 7	39.5	0.02	2.1E + 6	27.2	5.4E-3	2.4E + 6	25.3	6E-3
10	1.4E + 7	61.4	0.03	1.6E + 7	50.4	0.04	3.3E + 6	31.3	8.4E-3	3.7E + 6	28.3	9.5E-3
12	2E + 7	79.4	0.04	2.3E + 7	63.6	0.05	4.8E + 6	36.3	0.01	5.4E + 6	31.9	0.01
14	2.8E + 7	100	0.06	3.2E + 7	79.1	0.07	6.5E + 6	42.1	0.02	7.3E + 6	36.2	0.02
16	3.7E + 7	124	0.08	4.2E + 7	96.9	0.9	8.5E + 6	48.9	0.02	9.6E + 6	41.2	0.02
18	4.7E + 7	151	0.1	5.3E + 7	117	0.1	1E + 7	56.5	0.03	1.2E + 7	46.8	0.03
1.5 μm thickness							2 2μm thickness					
Ni_{50}-Ti_{50}				Ni_{40}-Ti_{50}-Cu_{10}			Ni_{50}-Ti_{50}			Ni_{40}-Ti_{50}-Cu_{10}		
I (mA)	Stress (N/m^2)	T (°C)	T.D (μm)	Stress (N/m^2)	T (°C)	T.D (μm)	Stress (N/m^2)	T (°C)	T.D (μm)	Stress (N/m^2)	T (°C)	T.D (μm)
2	5.5E+ 4	20.2	1.5E-4	6.2E + 4	20.2	1.7E-4	3E + 4	20.1	8.7E-5	3.3E + 4	20.1	9.8E-5
4	2.2E + 5	20.8	6.1E-4	2.5E + 5	20.6	6.9E-4	1.2E + 5	20.5	3.5E-4	1.3E + 5	20.3	3.9E-4
6	5E + 5	21.9	1.4E-3	5.6E + 5	21.4	1.6E-3	2.7E + 5	21.1	7.9E-4	3E + 5	20.8	8.9E-4
8	8.9E + 5	23.3	2.5E-3	10E + 5	22.4	2.8E-3	4.7E + 5	21.9	1.4E-3	5.3E + 5	21.4	1.6E-3
10	1.4E + 6	25.2	3.8E-3	1.6E + 6	23.8	4.3E-3	7.4E + 5	23	2.2E-3	8.3E + 5	22.2	2.5E-3
12	2E + 6	27.5	5.5E-3	2.3E + 6	25.5	6.2E-3	1E + 6	24.3	3.1E-3	1.2E + 6	23.1	3.5E-3
14	2.7E + 6	30.2	7.5E-3	3E + 6	27.4	8.5E-3	1.5E + 6	25.8	4.3E-3	1.6E + 6	24.2	4.8E-3
16	3.6E + 6	33.3	9.8E-3	4E + 6	29.7	0.01	1.9E + 6	27.6	5.6E-3	2.1E + 6	25.5	6.3E-3
18	4.5E + 6	36.8	0.01	5E + 6	32.3	0.01	2.4E + 6	29.6	7E-3	2.7E + 6	27	8E-3

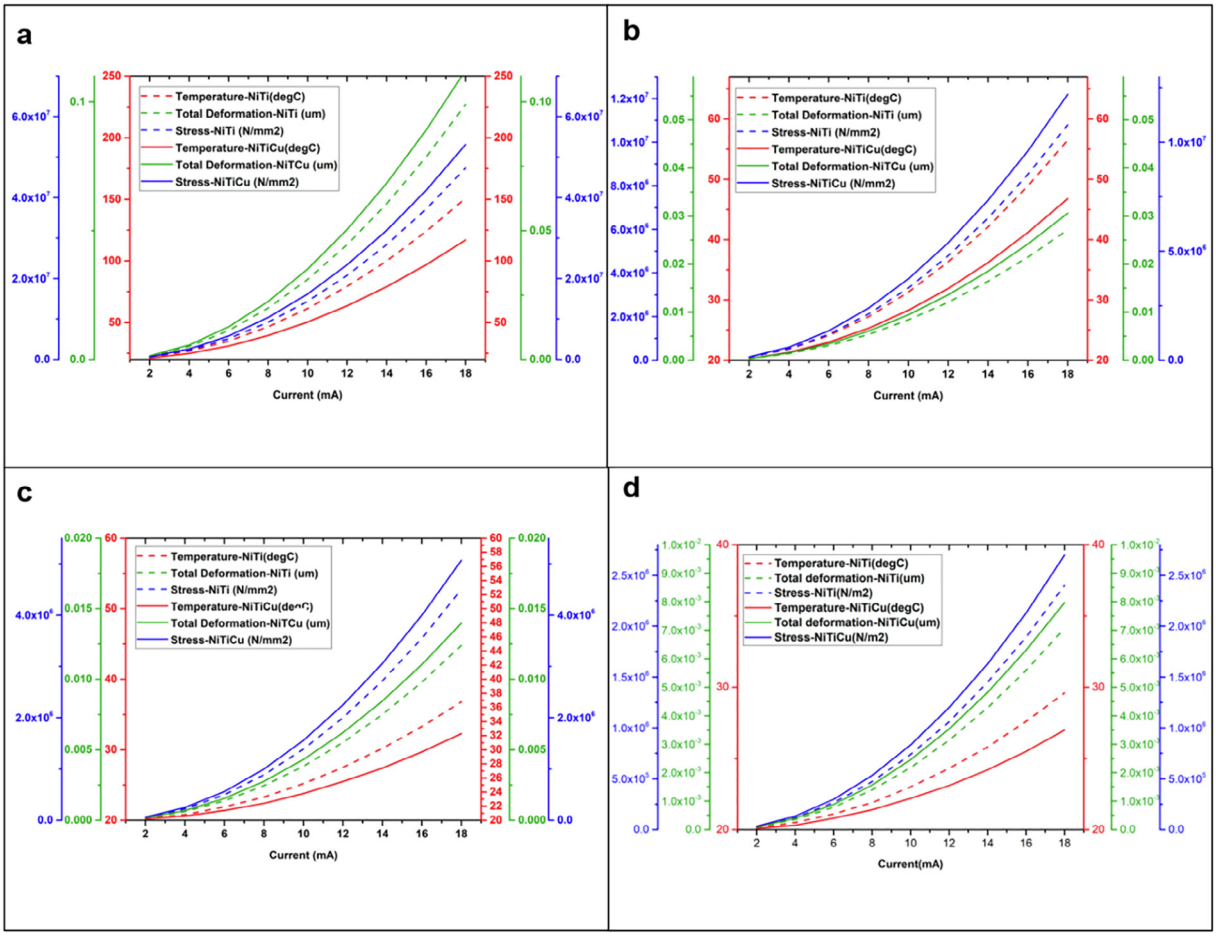


Fig. 8. Comparative analysis between Ni-Ti & Ni-Ti-Cu for different values of stress, deformation, and temperature depending upon the input current in (a) 0.5 μm, (b) 1 μm, (c) 1.5 μm, (d) 2 μm thickness film.

4.3. Comparative analysis

As per Fig. 8, it is found out from the analysis and simulations that the thermal stresses generated in Ni-Ti SMA are lower as compared with Ni-Ti-Cu for the same amount of input current. The temperature generated in the Ni-Ti material is more than Ni-Ti-Cu. Whereas, in the case of total displacement, it is found that Ni-Ti-Cu shows a higher value than of Ni-Ti. Therefore, it can be concluded from the above data that Ni-Ti-Cu shows better deformation behavior. Thus it will show better shape memory behavior than Ni-Ti when the same current is provided for Joule's heating actuation. Much more stress is generated in Ni-Ti-Cu than that of Ni-Ti material which may affect the fatigue behavior of Ni-Ti-Cu. The temperature generated in Ni-Ti-Cu is less, and still, it showed more deformation. Thus implicating that Ni-Ti-Cu can actuate better, requiring less heat.

Few researchers have tried to do a comparative analysis between Ni-Ti and Ni-Ti-Cu by conducting actuation applications. The results suggest that Ni-Ti-Cu showed better actuation than Ni-Ti [27,24]. Therefore, the results obtained from the present FE analysis are in agreement with the few experimental results presented in the pertinent literature.

5. Conclusion

The actuation behavior of the Ni_{50}-Ti_{50} and Ni_{40}-Ti_{50}-Cu_{10} SMA is simulated using the finite element analysis based on Joule's heating module in the COMSOL Multiphysics environment. The temperature, stress, and deformation increased with an increase in input current. The effect of varying thickness was analyzed and found that with an increase in film thickness, the deformation, the temperature generated, and the stress are decreased. The comparative analysis predicts that the Ni-Ti-Cu shows better actuation for the same value of input current.

CRediT authorship contribution statement

Anita Jena: Conceptualization, Methodology, Formal analysis, Writing - original draft. **Bijaya Bikram Samal:** Data curation, Formal analysis, Writing - review & editing. **Cheruvu Siva Kumar:** Supervision, Resources, Writing - review & editing. **Shailendra Kumar Varshney:** Supervision, Resources, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] J.G. Korvink, O. Paul, MEMS: A practical guide of design, analysis, and applications. 2006.

[2] W.C. Crone, "A Brief Introduction to MEMS and NEMS," in Springer Handbook of Experimental Solid Mechanics, W. N. Sharpe, Ed. Boston, MA: Springer US, 2008, pp. 203–228.

[3] J. M. Gallardo Fuentes, P. Gümpel, J. Strittmatter, "Phase Change Behavior of Nitinol Shape Memory Alloys," Advanced Engineering Materials, vol. 4, no. 7, pp. 437–452, Jul. 2002, 10.1002/1527-2648(20020717)4:7<437::AID-ADEM437>3.0.CO;2-8.

[4] C.R. Knick, "Fabrication and Characterization of Nanoscale Shape Memory Alloy MEMS Actuators," in Advanced Functional Materials [Working Title], 2020.

[5] Y. Fu, H. Du, W. Huang, S. Zhang, M. Hu, TiNi-based thin films in MEMS applications: a review, Sens. Actuators, A 112 (2) (2004) 395–408, https://doi.org/10.1016/j.sna.2004.02.019.

[6] S. Aramaki, S. Kaneko, K. Arai, Y. Takahashi, H. Adachi, K. Yanagisawa, "Tube type micro manipulator using shape memory alloy (SMA)," in MHS'95. Proceedings of the Sixth International Symposium on Micro Machine and Human Science, Oct. 1995, pp. 115–120, doi: 10.1109/MHS.1995.494227.

[7] A.-L. Zhang, X.-Q. Zhang, W.-Y. Hu, X.-T. Fu, A shape memory alloy microvalve switching off by surface acoustic wave, Ferroelectrics 506 (1) (2017) 1–9, https://doi.org/10.1080/00150193.2017.1281676.

[8] C. Megnin, J. Barth, M. Kohl, A bistable SMA microvalve for 3/2-way control, Sens. Actuators, A 188 (2012) 285–291, https://doi.org/10.1016/j.sna.2011.11.016.

[9] W.L. Benard, H. Kahn, A.H. Heuer, M.A. Huff, Thin-film shape-memory alloy actuated micropumps, J. Microelectromech. Syst. 7 (2) (1998) 245–251, https://doi.org/10.1109/84.679390.

[10] M. Kohl, K.D. Skrobanek, S. Miyazaki, Development of stress-optimised shape memory microvalves, Sens. Actuators, A 72 (3) (1999) 243–250, https://doi.org/10.1016/S0924-4247(98)00221-0.

[11] E. Makino, T. Mitsuya, T. Shibata, Micromachining of TiNi shape memory thin film for fabrication of micropump, Sens. Actuators, A 79 (3) (2000) 251–259, https://doi.org/10.1016/S0924-4247(99)00290-3.

[12] E. Makino, T. Mitsuya, T. Shibata, Fabrication of TiNi shape memory micropump, Sens. Actuators, A 88 (3) (2001) 256–262, https://doi.org/10.1016/S0924-4247(00)00522-7.

[13] J. Abadie, N. Chaillet, C. Lexcellent, Modeling of a new SMA micro-actuator for active endoscopy applications, Mechatronics 19 (4) (2009) 437–442, https://doi.org/10.1016/j.mechatronics.2008.11.010.

[14] Y. Fu, H. Du, Fabrication of micromachined TiNi-based microgripper with compliant structure, Smart Sensors, Actuators, and MEMS 5116 (2003) 38–47, https://doi.org/10.1117/12.498184.

[15] S. Takeuchi, I. Shimoyama, A three-dimensional shape memory alloy microelectrode with clipping structure for insect neural recording, J. Microelectromech. Syst. 9 (1) (2000) 24–31, https://doi.org/10.1109/84.825773.

[16] J.J. Gill, D.T. Chang, L.A. Momoda, G.P. Carman, Manufacturing issues of thin film NiTi microwrapper, Sens. Actuators, A 93 (2) (2001) 148–156, https://doi.org/10.1016/S0924-4247(01)00646-X.

[17] A.P. Lee, D.R. Ciarlo, P.A. Krulevitch, S. Lehew, J. Trevino, M.A. Northrup, A practical microgripper by fine alignment, eutectic bonding and SMA actuation, Sens. Actuators, A 54 (1) (1996) 755–759, https://doi.org/10.1016/S0924-4247(97)80052-0.

[18] A. AbuZaiter, M. Nafea, A.A. Mohd Faudzi, S. Kazi, M.S. Mohamed Ali, "Thermomechanical behavior of bulk NiTi shape-memory-alloy microactuators based on bimorph actuation," Microsystem Technologies, vol. 22, no. 8, Aug. 2016, doi: 10.1007/s00542-015-2641-1.

[19] W. Huang, J.P. Tan, X. Gao, J.H. Yeo, Design, testing, and simulation of NiTi shape-memory-alloy thin-film-based microgrippers, J. Micro/Nanolithogr. MEMS MOEMS 2 (3) (2003) 185–190, https://doi.org/10.1117/1.1585063.

[20] M. Geetha, K. Dhanalakshmi, S. Jayachandran, I.A. Palani, V. Sathish Kumar, Analysis of actuation characteristics of CuAlNiMn/Polyimide thin film shape memory alloy, Mater. Today: Proc. (2020), https://doi.org/10.1016/j.matpr.2020.04.689.

[21] D.C. Lagoudas, A. Bhattacharyya, Modeling of thin layer extensional thermoelectric SMA actuators, Int. J. Solids Struct. 35 (3) (1998) 331–362, https://doi.org/10.1016/S0020-7683(97)00067-X.

[22] C. Velez et al., Rapid Prototyping of Microactuators by Integrating 3D Printed Polymeric Structures with NiTi Thin Film, in 2020 IEEE 33rd International Conference on Micro Electro Mechanical Systems (MEMS), Jan. 2020, pp. 893–896, doi: 10.1109/MEMS46641.2020.9056231.

[23] D.J. Hartl, D.C. Lagoudas, "Thermomechanical Characterization of Shape Memory Alloy Materials," in Shape Memory Alloys, Boston, MA: Springer US, 2008.

[24] S. Jayachandran et al., "Investigations on performance viability of NiTi, NiTiCu, CuAlNi and CuAlNiMn shape memory alloy/Kapton composite thin film for actuator application," Composites Part B: Engineering, vol. 176, p. 107182, 2019, 10.1016/j.compositesb.2019.107182.

[25] M. Thomasová et al., Evolution of macroscopic elastic moduli of martensitic polycrystalline NiTi and NiTiCu shape memory alloys with pseudoplastic straining, Acta Mater. 123 (2017) 146–156, https://doi.org/10.1016/j.actamat.2016.10.024.

[26] W. Huang, On the selection of shape memory alloys for actuators, Mater. Des. 23 (1) (2002) 11–19, https://doi.org/10.1016/S0261-3069(01)00039-5.

[27] C. Bechtold, C. Chluba, C. Zamponi, E. Quandt, R.L. de Miranda, Fabrication and characterization of freestanding NiTi based thin film materials for shape memory micro-actuator applications, Shape Memory and Superelasticity 5 (4) (2019) 327–335, https://doi.org/10.1007/s40830-019-00239-0.